Personal details and date of CV

- **Surname:** Liu
- **First names:** Zhuchenyang
- **Researcher ID (ORCID):** 0009-0003-5034-2551
- **Links:** LinkedIn Profile
- **Date of CV:** October 22, 2026

Degree

- **2025-06-18: Master of Science in Data Science (Dual Degree)**
  - KTH Royal Institute of Technology (Sweden) & Aalto University (Finland) (EIT Digital Master School).
  - *Major:* Data Science. *GPA:* 4.56/5.0, Pass with Honours.
  - Achieved maximum grades in all technical courses.
- **2023-06: Bachelor of Science**
  - Jiangxi University of Finance and Economics (China).
  - *Major:* Data Science and Big Data Technology.

Current employment

- **2025-09 – Present: Doctoral Researcher (Fixed-term)**
  - **Employer:** Aalto University, Department of Information and Communications Engineering, Espoo, Finland.
  - **Supervisor:** Prof. Yu Xiao.
  - **Dissertation Topic:** Toward Efficient Visual Document Retrieval with Index Compression and Model Distillation.
  - **Doctoral progress:** Passed the mid-term review on 02.10.2026 (PhD Candidate). Expected graduation: Q4 2027.
  - **Key Project (AiWo):** Lead Researcher for NLP and Instruction Generation, collaborating with Finnish industrial partners. Designed a decoupled Multimodal Retrieval-Augmented Generation (Visual RAG) framework using ColPali. Successfully implemented an end-to-end local prototype for indexing and inference.
  - **Stage of academic research career:** Stage I (Doctoral Candidate).

Previous work experience

- **2025-02 – 2025-08: Master Thesis Intern**
  - **Employer:** ABB Corporate Research, Västerås, Sweden.
  - *Description:* Developed "AcademicRAG", a Graph-based RAG framework. Proposed a novel clues-based retrieval algorithm utilizing Knowledge Graphs and LLMs, outperforming SOTA baselines in academic Q&A benchmarks.
- **2022-06 – 2022-08: System Architecture Intern**
  - **Employer:** Whale Cloud Technology Co., Ltd, China.
  - *Description:* Assisted in designing high-level architecture for multimodal search functions in a smart city intelligent system.

Research output

Publications

- **1. Liu, Z.**, Zhang, Y. & Xiao, Y. (2026). *NanoVDR: Distilling a 2B Vision-Language Retriever into a 70M Text-Only Encoder for Visual Document Retrieval*. arXiv preprint arXiv:2603.12824. (Accepted to EMNLP 2026, Main Conference). [GitHub] [HuggingFace]

  - *Role:* First author. Proposed an asymmetric distillation framework achieving 95% teacher quality with 32× fewer parameters.
- **2. Liu, Z.**, Hu, Z., Zhang, Y. & Xiao, Y. (2026). *Structural Anchor Pruning: Training-Free Multi-Vector Compression for Visual Document Retrieval*. arXiv preprint arXiv:2601.20107. (Accepted to EMNLP 2026, Main Conference). [GitHub]
  - *Role:* First author. Conceived the structural anchor pruning method to optimize Visual RAG indexing efficiency.
- **3. Liu, Z.**, Wang, Z., Zhang, Y. & Xiao, Y. (2026). *DistilVDR: A Compact End-to-End Visual Document Retriever via Dual-Student Distillation*. arXiv preprint arXiv:2608.10636. (Accepted to DocInsights 2026: Workshop on Document Intelligence and Understanding, co-located with EMNLP 2026, Budapest, Hungary). [GitHub] [HuggingFace]
  - *Role:* First author. Distilled a 524M end-to-end retriever from an 8B teacher, retaining ∼87% quality with a 15× smaller index.
- **4. Liu, Z.**, Wang, Z., Zhang, Y. & Xiao, Y. (2026). *ColNanoVDR: Document-Free Query Distillation for Multi-Vector Visual Document Retrieval via Optimal Transport*. arXiv preprint arXiv:2609.34899. (Under Review). [GitHub] [Demo] [Models]
  - *Role:* First author. Proposed document-free query distillation via optimal transport, retaining ∼95% teacher quality at 26× faster query encoding.
- **5. Liu, Z.**, Zhang, Y. & Xiao, Y. (2026). *Inverting Multi-Vector Visual Document Indices*. arXiv preprint arXiv:2610.09920. (Under Review).
  - *Role:* First author. Showed that stored multi-vector retrieval indices can be inverted into readable document pages, and identified which index-side defences actually hold; code and model weights withheld for responsible disclosure.
- **6. Liu, Z.**, Zhang, Y. & Xiao, Y. (2026). *Multimodal Reranking Is a System One Decision*. arXiv preprint arXiv:2610.28435. (Position paper). [Project Page] [GitHub]
  - *Role:* First author. Argued that multimodal reranking is better posed as a single multiple-choice decision over the candidate list than as per-candidate relevance scoring.
- **7. Liu, Z.**, Zhang, Y. & Xiao, Y. (2026). *Evidence Attribution in Visual Document Understanding without Coordinates or Region Labels*. arXiv preprint arXiv:2607.24651. (Under Review). [GitHub]
  - *Role:* First author. Proposed a language-cue attribution interface (quote-and-retrieve) and identified coordinate output, not localization ability, as the grounding bottleneck.
- **8. Liu, Z.**, Zhang, Y. & Xiao, Y. (2026). *IKEA-Bench: Benchmarking and Mechanistic Analysis of Vision-Language Models for Cross-Depiction Assembly Instruction Alignment*. arXiv preprint arXiv:2604.00913. (Under Review). [Project Page]
  - *Role:* First author. Constructed the benchmark and conducted the three-layer mechanistic analysis of VLMs.
- **9. Liu, Z.**, Zhang, Y., He, Y., Paasio, H., Li, C., Semjonova, G. & Xiao, Y. (2026). *Exploring Human-AI Collaboration in E-Textile Design: A Case Study on Flex Sensor Placement for Shoulder Motion Detection*. Proceedings of the ACM on Human-Computer Interaction, Vol. 10, No. EICS, Article EICS019. (Published; oral presentation at EICS 2026, Greece). doi:10.1145/3816771
  - *Role:* First author. Investigated LLM-assisted sensor placement optimization for e-textile motion capture.
- **10. Liu, Z.**, Qu, N., Zhang, Y., O'Mahony, M., Wang, Q. & Xiao, Y. (2026). *Mapping E-textiles Design Pain Points and Generative AI Opportunities: Insights from Workshops in Shanghai and Winchester*. arXiv preprint arXiv:2610.07296. (Accepted to the 8th International Conference on the Challenges, Opportunities, Innovations and Applications in Electronic Textiles, E-Textiles 2026).
  - *Role:* First author. Mapped e-textile design bottlenecks across two international

co-design workshops and identified where generative AI can realistically help.

- **11.** Zhang, Y., **Liu, Z.**, Ploetz, T. & Xiao, Y. (2026). *Encoder-Free Human Motion Understanding via Structured Motion Descriptions*. arXiv preprint arXiv:2604.21668. (Under Review).
  - *Role:* Co-author. Contributed to the structured motion description framework and LLM-based motion reasoning.
- **12.** Zhang, Y., **Liu, Z.**, He, Y., Ploetz, T. & Xiao, Y. (2026). *Fine-grained Motion Retrieval via Joint-Angle Motion Images and Token-Patch Late Interaction*. arXiv preprint arXiv:2603.09930. (Accepted to ACCV 2026).
  - *Role:* Co-author. Contributed to the late interaction retrieval framework and experimental design.
- **13.** Zhang, Y., **Liu, Z.** & Xiao, Y. (2026). *LingoMotion: An Interpretable and Unambiguous Symbolic Representation for Human Motion*. In 2026 IEEE International Conference on Pervasive Computing and Communications (PerCom), Work-in-Progress Track. (Accepted).
  - *Role:* Co-author. Contributed to the design of the symbolic representation framework and experimental validation.
- **14.** Yang, Z., **Liu, Z.**, Zhang, Y., Leite, C. & Xiao, Y. (2026). *StrengthSense: A Dataset of IMU Signals Capturing Everyday Activities with Diverse Physical Demands*. In Proceedings of the 2026 ACM International Symposium on Wearable Computers (ISWC '26), pp. 209–215. doi:10.1145/3830727.3834811
  - *Role:* Co-author. Contributed to the dataset design and the evaluation of IMU-based activity recognition across levels of physical demand.
- **15.** Eloranta, A., **Liu, Z.**, Naulapaa, I., Arvola, I., Zhang, Y., Leppisaari, A.-M., Xu, L. & Xiao, Y. (2026). *Knit-Structure Effects on Electromechanical Metrics and Their Correlation with Joint-Angle Estimation Error in Knitted Strain Sensors*. arXiv preprint arXiv:2610.07416. (Under Review).
  - *Role:* Co-author. Contributed to the evaluation protocol linking knit structure to joint-angle estimation error.
- **16. Liu, Z.**, Chen, S., Tavallaey, S.S., Heintz, F. & Zea, E. (2025). *AcademicRAG: A Knowledge Graph-Enhanced Framework for Intelligent Academic Resource Discovery*. Big Data Analytics & Applications, p.5.
  - *Role:* First author. Led the development of the KG-enhanced RAG framework during Master's thesis at ABB.
- **17. Liu, Z.** (2022). *User Adoption Analysis of Social Media Applications Based on NLP and Technology Acceptance Model*. 2022 5th International Conference on Data Science and Information Technology (IEEE-DSIT).
  - *Role:* Sole author.

Other research outputs

- **Benchmark contributions (MTEB):** Contributed model integrations and evaluation results to the Massive Text Embedding Benchmark (MTEB): added `NanoVDR-S-Multi`, including a custom asymmetric encoder interface for visual document retrieval, and the `ColNanoVDR` late-interaction query towers, together with their ViDoRe v1/v2/v3 results on the public leaderboard. Four merged pull requests across `embeddings-benchmark/mteb` and `embeddings-benchmark/results` (2026).
- **Software/Tool:** AcademicRAG (GitHub) - A comprehensive framework integrating Knowledge Graphs with RAG for academic literature retrieval.

Awards and honours

- **2025: Kaggle Competitions Master**
  - Awarded the Master tier (Ranked top 0.5% of the global data science community) for

consistent high-performance in international machine learning competitions.
  - *Granting body:* Kaggle (Google).
- **2023: Overseas Study Scholarship for Outstanding Undergraduates**
  - Awarded for academic excellence during Bachelor's studies.
  - *Granting body:* Jiangxi University of Finance and Economics.

Other expertise of relevance

- **Technical Skills:**
  - *Core:* Python (PyTorch), LLM Fine-tuning, RAG Pipelines, Knowledge Graphs.
  - *Tools:* Docker, Git, Linux, HuggingFace Ecosystem.